

SAYLANI MASS IT TRAINING (SMIT)

Data Science & Artificial Intelligence — Assignment
Machine Learning Project: Loan Approval Prediction

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1. EXECUTIVE SUMMARY

The objective of this project is to build a robust Predictive Machine Learning model that automates the loan approval process for financial institutions. Manual processing of loan applications is time-consuming, prone to human error, and highly vulnerable to subjective bias. By utilizing applicant demographic details and financial histories, this project evaluates three distinct classification algorithms: Logistic Regression, Decision Tree Classifier, and Random Forest Classifier. Data preprocessing techniques, including missing value imputation, categorical encoding, feature scaling, and class-balancing via SMOTE, were implemented. The evaluation metrics demonstrate that Logistic Regression is the most optimal model, achieving an overall accuracy of 76% and an F1-score of 0.83 for approved cases, establishing a reliable data-driven mechanism for credit risk management.

2. PART 1: DATA UNDERSTANDING & FEATURE DESCRIPTION

The dataset contains historical records of loan applicants, capturing their economic standing, demographic profile, and historical credit behavior. Shape: 614 rows and 13 columns. Target Variable: Loan_Status (Categorical: Y = Approved, N = Rejected).

Feature Name	Data Type	Description
Loan_ID	Object / String	Unique alphanumeric identifier for each application.
Gender	Categorical	Gender of the applicant (Male / Female).
Married	Categorical	Marital status of the applicant (Yes / No).
Dependents	Categorical	Number of dependent family members (0, 1, 2, 3+).
Education	Categorical	Educational qualification status (Graduate / Under Graduate).
Self_Employed	Categorical	Employment status type (Yes / No).
ApplicantIncome	Numerical (Int)	Total monthly income of the primary applicant.
CoapplicantIncome	Numerical (Float)	Monthly income contributed by the co-applicant.

LoanAmount	Numerical (Float)	Requested loan principal amount (in thousands).
Loan_Amount_Term	Numerical (Float)	Repayment duration period in months.
Credit_History	Numerical (Binary)	Credit score compliance record (1.0 = Good, 0.0 = Bad).
Property_Area	Categorical	Location type of applicant's property (Urban / Semiurban / Rural).
Loan_Status	Categorical	Target Class indicating loan approval verdict (Y / N).

3. PART 2: DATA PREPROCESSING PIPELINE

- **Handling Missing Values:** Null values were identified in multiple variables. Categorical columns (Gender, Married, Dependents, Self_Employed) were filled using the Mode of their respective columns. Numerical features (LoanAmount, Loan_Amount_Term) were imputed using the Median to mitigate the effect of outliers. Credit_History was treated using the mode due to its binary nature.
- **Feature Encoding:** Label Encoding was applied to binary targets like Loan_Status (Y->1, N->0), Gender, and Married. One-Hot Encoding was utilized for multi-class categorical features such as Property_Area and Dependents to avoid artificial ordinal rankings.
- **Feature Scaling:** Because numerical values like ApplicantIncome and LoanAmount operate on drastically different mathematical scales, StandardScaler was applied to normalize numerical vectors to a mean of 0 and a variance of 1.
- **Handling Class Imbalance:** The original dataset exhibited a heavy skew towards approved loans. SMOTE (Synthetic Minority Over-sampling Technique) was deployed exclusively on the training subset to synthesize balanced instances of the minority class (Rejected), ensuring the models did not build a prediction bias.

4. PART 3: EXPLORATORY DATA ANALYSIS (EDA) INSIGHTS

- **Insight 1: Credit History Impact:** A cross-tabulation analysis reveals that applicants possessing a positive Credit_History (1.0) have an exponentially higher probability of loan approval. This stands out as the single highest correlated independent variable.
- **Insight 2: Distribution of Applicant Income:** The distribution plot displays a heavy right-skewed distribution. Most applicants fall under the lower-to-middle monthly income bracket, with extreme outliers pulling the tail to the right.
- **Insight 3: Distribution of Loan Amount:** Similar to income, requested loan amounts are highly concentrated within lower parameters, showing a steady bell-shaped distribution that flattens out around the higher credit brackets.

- **Insight 4: Loan Status Class Proportions:** Initial target analysis showed approximately 68% approved cases versus 32% rejected applications, validating the structural necessity of SMOTE balancing prior to training.
- **Insight 5: Correlation Heatmap:** Pearson’s correlation matrix highlights a moderate positive correlation between ApplicantIncome and LoanAmount. This proves that as individual income scales up, institutions naturally tolerate higher loan risk capacities.

5. PART 4 & 5: MODEL BUILDING & EVALUATION

1. Logistic Regression Classifier (Accuracy: 76%)

Class	Precision	Recall	F1-Score	Support
0 (Rejected)	0.73	0.51	0.60	43
1 (Approved)	0.77	0.90	0.83	80

Confusion Matrix Outcomes: True Negatives: 22 | False Positives: 21 | False Negatives: 8 | True Positives: 72

2. Decision Tree Classifier (Accuracy: 70%)

Class	Precision	Recall	F1-Score	Support
0 (Rejected)	0.57	0.56	0.56	43
1 (Approved)	0.77	0.78	0.77	80

Confusion Matrix Outcomes: True Negatives: 24 | False Positives: 19 | False Negatives: 18 | True Positives: 62

3. Random Forest Classifier (Accuracy: 71%)

Class	Precision	Recall	F1-Score	Support
0 (Rejected)	0.61	0.44	0.51	43
1 (Approved)	0.74	0.85	0.79	80

Confusion Matrix Outcomes: True Negatives: 19 | False Positives: 24 | False Negatives: 12 | True Positives: 68

6. PART 6: COMPARATIVE ANALYSIS & MODEL SELECTION

Classifier Model	Overall Accuracy	Class 1 Recall	Class 1 F1-Score	False Negatives
Logistic Regression	76% (Highest)	90% (Highest)	0.83 (Highest)	8 (Lowest)
Decision Tree	70%	78%	0.77	18
Random Forest	71%	85%	0.79	12

- Performance Breakdown:
- 1. Accuracy Supremacy: Logistic Regression achieves the maximum overall test accuracy of 76%.
 - 2. Error Minimization: In credit underwriting, a False Negative means rejecting a reliable applicant. Logistic Regression dropped this error to just 8 occurrences.
 - 3. Sensitivity Metrics: Logistic Regression demonstrates an elite 90% Recall on Class 1.

7. BUSINESS RECOMMENDATIONS & CONCLUSION

By transforming unstructured demographic markers into mathematically scaled numerical profiles and engineering a clean predictive setup, this machine learning project successfully automates credit decision workflows. Logistic Regression stands out as the optimal algorithmic solution for implementation at SMIT, ensuring minimized business losses while driving swift and unbiased automated credit validations.

Hence, Logistic Regression is confidently selected as the optimal model for the automated loan approval prediction system.